2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) | 979-8-3315-8618-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/EMBC58623.2025.11253685

Multi-resolution and Multi-modal Feature Integration using Graph Neural Networks for Optical Coherence Tomography Image Classification

Nethmi Dissanayake[1], Tharindu Fernando[2], Aveesha Gunasekara[1], Sridha Sridharan[2], and Clinton Fookes[2]

***Abstract*— Early and accurate diagnosis of retinal diseases is crucial, as many of these conditions can be prevented or managed effectively with timely treatment. Optical Coherence Tomography (OCT) is a widely used medical imaging technique that provides high-resolution visualization of the retina, allowing the detection of abnormalities. Current state-of-the-art methods predominantly rely on extracting either local or global features using Convolutional Neural Networks or Vision Transformers from OCT images. Moreover, few existing ensemble methods primarily focus on concatenation and decision-level feature fusion, limiting their ability to fully capture the relationships between distinct feature types. In this study, we propose to incorporate multi-modal semantic features extracted at multiple feature resolutions to achieve better classification of OCT images. In addition, to effectively fuse the multitude of extracted information, a learnable graph-based effective feature fusion method is proposed. This allows us to effectively capture complex lesion patterns and subtle abnormalities in the retina. We evaluate our method using three publicly available datasets: OCTDL, OCTID, and OCT2014 and surpass the performance of current state-of-the-art models with significant margins, demonstrating the robustness and reliability of our approach for retinal disease classification.**

***Keywords*— Retinal Image Classification; Optical Coherence Tomography; Multi-Resolution Feature Fusion; Model Ensembles; Graph Neural Networks.**

I. INTRODUCTION

According to statistics from the World Health Organization (WHO), approximately 2.2 million people worldwide experience vision impairment, most of which could have been prevented with early detection [1]. Retinal diseases are among the most common causes of vision loss, and hence the early and accurate diagnosis of retinal diseases is essential in preventing temporary or permanent vision loss and vision abnormalities [2], [3].

Optical Coherence Tomography (OCT) has become a vital, non-invasive imaging technique in ophthalmology, providing high-resolution, cross-sectional retinal images using light waves, allowing detailed visualisation of the distinctive layers of the optic nerve and the retina [4]. Manually analysing OCT images is time-consuming and susceptible to human errors, posing significant challenges to a timely and accurate diagnosis [5]. This issue is further compounded by a global shortage of ophthalmologists [6]. As such, automated systems that can accurately analyze OCT scans and recognise retinal diseases are essential for ensuring timely diagnosis and effective treatment, particularly in remote and developing regions where access to specialised healthcare is limited.

The tremendous success of Deep Learning (DL) methods has opened new pathways for the automated diagnosis of retinal diseases. Especially, Convolutional Neural Networks (CNNs) have shown remarkable potential in extracting meaningful attributes from medical images. Several studies have adapted some pre-existing CNN architectures for retinal disease classification using OCT images [7], [8], [9], while some studies have implemented custom CNN models [10], [11]. Additionally, [12] proposes a Coherent Convolutional Neural Network (CCNN) to ensure consistent feature propagation between layers and effectively detect irregular patterns in retinal diseases.

Although CNNs are highly effective in capturing local features and provide well-informed inductive biases, they often struggle to capture the global context [13]. On the other hand, Vision Transformers (ViTs) excel at extracting the global features and long-range dependencies among image pixels through its self-attention mechanism [14] but lacks the ability to extract the fine-grained local information. Extraction of both local and global contextual features is important for accurate OCT diagnosis. For example, the local features explain the lesion characteristics in the retina, while global semantics could explain complex associations that they have among the different components in the retina. Combining all of this information can provide a more comprehensive representation of the retinal structure.

Recent studies have demonstrated the application of ensemble models to integrate various retinal features extracted from distinct DL models. Specifically, feature-level fusion methodologies [15], [16], [17], decision-level ensembles such as soft voting ensembles [18], [19], bagging ensembles [20], and dynamic weighting strategies [21] are among the most common approaches found in literature in ensemble-based OCT image analysing methods. In addition to these ensemble-based approaches, multi-resolution features derived from the same CNN backbone feature extractor has been recently utilised in [22], [23] identifying both fine-grained local features and global contextual information.

While ensemble-based methods have shown promise compared to their single-model counterparts, we argue that the full potential of ensemble models in OCT image analysis is

[1]N. Dissanayake and A. Gunasekara are with the Faculty of Engineering, University of Peradeniya, Sri Lanka. (e17075, e17098)@eng.pdn.ac.lk

[2]T. Fernando, S. Sridharan, and C. Fookes are with the SAIVT Research lab, Queensland University of Technology, Brisbane, Australia. (t.warnakulasuriya, s.sridharan, c.fookes).qut.edu.au

yet to be discovered. In particular, we observe that different pre-trained CNN-based backbones leveraged in current literature are able to capture different, fine-grained, local semantics of the input OCT image, therefore, by extracting features from multiple layers of the all the CNN-based models that encompass the model ensemble could provide local semantic information at multiple levels of abstraction. Furthermore, we propose to augment this extracted fine-grained information with the global contextual information of the input OCT image that a ViT model could extract. Moreover, we propose a novel graph neural network based fusion strategy to fuse this extracted information effectively by learning the dynamics across features extracted at multiple levels of the same model as well as features of different models. In addition to learning the dynamics, the graph-based fusion strategy allows us to fuse the features efficiently by maintaining the learnable parameters at a tractable level, which is highly beneficial due to the labeled data scarcity in OCT image analysis. While we acknowledge the existence of architecture for graph-based fusion of features from the same or different models, to the best of our knowledge, this is the first work to propose the fusion of multi-resolution multi-model features using a graph neural network. The main contributions of the proposed work can be summarised as follows:

1) We propose a novel model ensemble-based deep learning architecture for OCT image analysis.
2) We illustrate the utility of multi-resolution feature extraction and the merging of local and global contexts for effective diagnosis.
3) We introduce a robust and efficient graph neural network-based fusion strategy for fusing the extracted multi-resolution multi-model features
4) We demonstrate the effectiveness of the proposed framework using multiple public benchmarks in which our model outperforms the state-of-the-art methods.

The rest of this paper is organised as follows: Section II describes the dataset, and Section III addresses our proposed methods for classifying OCT images. Section IV discusses the evaluation results, followed by the conclusion in Section V.

II. DATASETS

In this study, we used three publicly available datasets, the Optical Coherence Tomography Dataset for Image-Based Deep Learning Methods (OCTDL) [24], the Optical Coherence Tomography Image Database (OCTID) [25], and the Optical Coherence Tomography - 2014 (OCT-2014) [26] datasets.

A. OCTDL Dataset

This dataset comprises over 2000 high-resolution OCT-B scans with various diseases and eye conditions that allow the visualization of the retinal layers on the fovea, posterior vitreous body, and choroidal vessels [24]. The image classes include Age-related Macular Degeneration (AMD), Diabetic Macular Edema (DME), Epiretinal Membrane (ERM), Normal (NO) Retinal Artery Occlusion (RAO), Retinal Vein Occlusion (RVO), and Vitreomacular Interface Disease (VID). Following prior work, this dataset has been divided into training, validation, and testing splits, with the proportion of 60:10:30 at the patient level, respectively, ensuring one patient is included in only one subset.

B. OCTID Dataset

The OCTID dataset consists of over 500 high-resolution OCT images categorised by various pathological conditions, including AMD, Central Serous Retinopathy (CSR), Diabetic Retinopathy (DR), Macular Holes (MH), and normal eyes (NO) [27]. Following prior work, this dataset has been randomly split into 70:10:20 splits for training, validation, and testing, respectively.

C. OCT-2014 Dataset

This dataset consists of volumetric scans collected from 15 healthy individuals, 15 AMD patients, and 15 DME patients. In total, this dataset has 723 AMD, 1101 DME, and 1407 normal OCT B-scan images at varying resolutions [26]. For training, validation, and testing, we used the splits provided by the dataset authors.

III. METHODS

The proposed framework is visually illustrated in Fig. 1, which is composed of: i) Multi-Resolution and Multi-Model Feature Extraction module, ii) Task-Specific Topology Generation and Graph-based Feature Fusion Module, iii) Edge Feature Enhancement Module, and iv) a Classifier. The structure and functionality of each of these modules are explained in the following subsections.

A. Multi-Resolution and Multi-Model Feature Extraction

Based on the prior works [28], [29], which have demonstrated the utility of using pre-trained ResNet50 [30] and Inception V3 [31] models for extracting meaningful features from retinal images, we leverage ResNet50 and Inception V3 as our CNN based backbones for feature extraction. In addition to the kernel-based features extraction, we used Vision Transformer (ViT) [32] to capture global contextual features in input OCT images. All the models are pre-trained using the ImageNet [33] datasets.

For the ResNet50 model, we utilized the feature vector from the final average pooling (AvgPool) layer, as well as from ”conv2_x” and “conv4_x” block outputs [30], with a Global Average Pooling (GAP) operation applied to the latter. In the Inception V3 model, we obtained features from the final average pooling layer and ”Mixed_6e” layer output by applying GAP. Additionally, we incorporated features from the ViT-Large-Patch 16-224 [34] model, specifically from the final encoding layer, and Layer 16 for capturing the global context.

Formally, this process can be written as:

$$X_i^{m,l} = f^{m,l}(X_i), \tag{1}$$

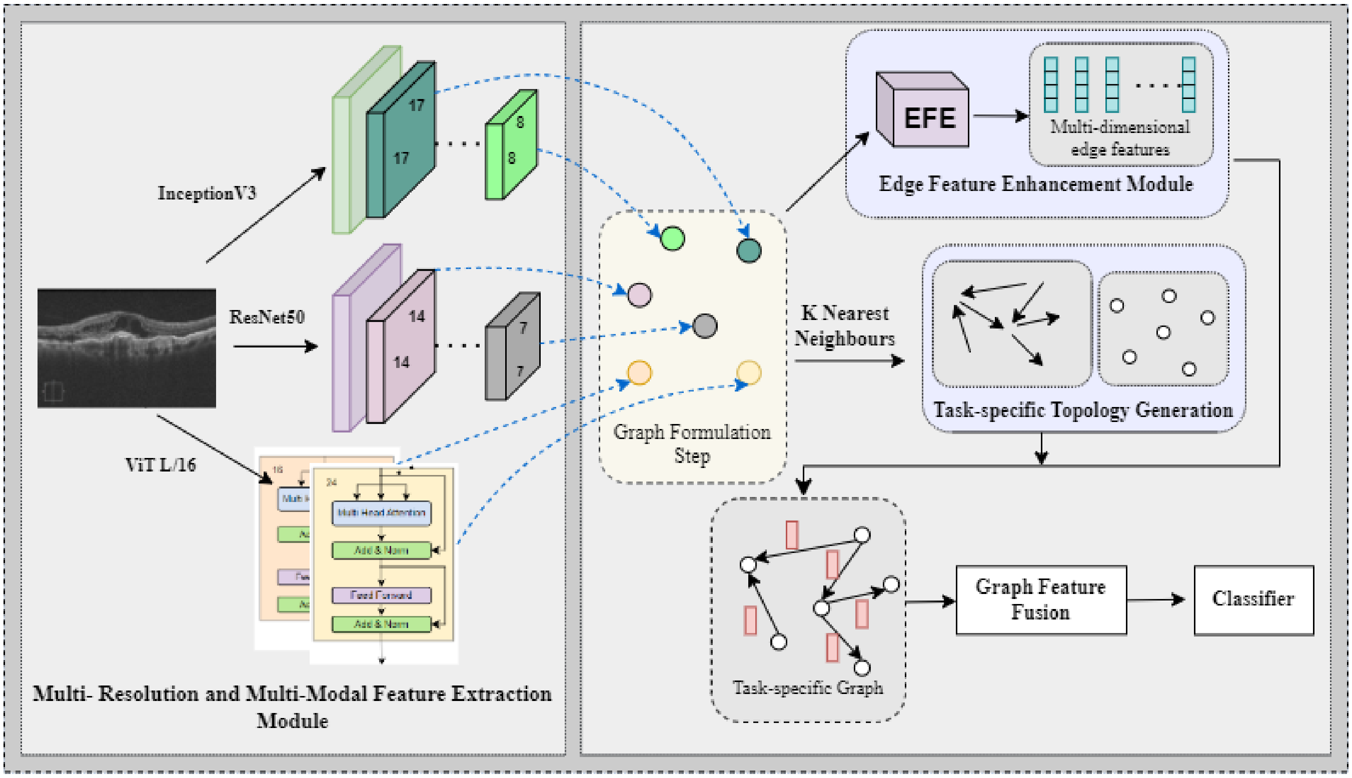

Fig. 1. Overall architecture of the proposed method which includes, (I) Muti-Resolution and Multi-Model Feature Extraction module which extracts local and global feature representations using pre-trained ResNet50, Inception V3, and ViT models from OCT images (See Sec. III-A), (II) A graph formulation step in which a basic graph-based structure is constructed with image feature vectors as nodal parameters. (III) Edge Feature Enhancement Module (EFE) module to generate multi-dimensional edge parameters (See Sec. III-C), (IV) Task-Specific Topology Generation (TTG) Module to dynamically generate and update the graph structure (See Sec. III-B), (V) Graph-based Feature Fusion module to generate the augmented feature vector based on EFE and TTG (See Sec. III-B), and (VI) A classifier to generate the predictions (See Sec. III-D).

where X_i denotes the input OCT image, m denotes the pre-trained feature extraction backbone, and l denotes the layer from which the features are extracted. The resultant feature vector $X_i^{m,l}$ is used as the feature representation of model m at resolution l. Using this procedure, we extract N feature embeddings.

The extracted feature embeddings are then fed into a transformer block to learn the correlation vectors, Z_i, across the extracted feature embeddings. These vectors capture the dependencies between across distinct feature vectors as:

$$Z_i = f^{T_{block}}(X_i^{m,l}), \tag{2}$$

where, $Z_i \in \mathbb{R}^{N \times D}$, comprises N number of refined feature vectors, each represented in D-dimensional space. Subsequently, the correlation matrix Z is passed through a Fully Connected (FC) layer, transforming it into $Q = \{q_1, q_2, \ldots, q_N\}$, the vertex feature matrix for the graph neural network that comprises N vertices. Then we leverage a graph neural network to fuse the information in the matrix Q into one unified feature vector. The correlation matrix is in Sec. III-B for the generation of task-specific graph neural network topologies.

B. Task-Specific Topology Generation (TTG) and Graph-based Feature Fusion Module

A graph neural network, G, can be represented as $G^A(Q, \varepsilon)$ where Q denotes the vertex feature matrix, ε denotes the edge features and A represents the adjacency matrix.

The adjacency matrix, A, plays a crucial role in the fusion of features as it guides the information flows between nodes. While the manual definition of A is possible using a predefined rule such as weighting based on the importance of a particular model or a resolution, this is sub-optimal as it depends on our interpretation of the features. Inspired by [35], we propose to generate a task-specific adjacency matrix, $\hat{A}$, as,

$$[\hat{Q}, \hat{A}] = f^{TTG}(Z, A, Q), \tag{3}$$

where, $\hat{A}$ denotes the task-specific adjacency matrix and $\hat{Q}$ denotes the task-specific vertex features.

The task-specific topology generation function, f^{TTG}, is governed by a rule that determines the presence of an edge between two feature vectors. The edge feature for each connected pair of vertices is 1, which can be expressed as:

$$\hat{\varepsilon} \subseteq \{e_{i,j} = 1 \mid \hat{q}_i, \hat{q}_j \in \hat{Q} \text{ and } \hat{A}_{i,j} = 1\}. \tag{4}$$

The task-specific adjacency matrix $\hat{A}$ is then defined as,

$$\hat{A}_{ij} = \begin{cases} 1 & \text{if } \hat{q}_j \in K(q_i) \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where, $K(q_i)$ denotes the K nearest neighbor vertices of vector q_i.

To further improve the vertex features and optimize the inter-relationships between vertices, we employ a Graph Convolutional Network (GCN) layer. The GCN jointly updates all vertex features based on the generated graph structure, leveraging $\hat{A}$ to learn optimal interactions between each model or each resolution. This can be expressed as,

$$G^A_{opt}(Q_{opt}, \varepsilon_{opt}) = f^{GCN}(G^A(Q, \varepsilon)) \tag{6}$$

where, f^{GCN} denotes the graph convolution operation, and Q_{opt}, and ε_{opt} represent node features and edge relationships respectively. This graph effectively encodes information about multiple retinal diseases, capturing relationships between local and global retinal features. The next step involves generating the fused feature vector for precise retinal disease classification. To achieve this, we apply multiple pooling operations to node features to obtain a single graph-level feature embedding represented by,

$$Q_f = Concat\left(\max_{i \in N} Q_{opt}(i), \sum_{i \in N} Q_{opt}(i), \frac{1}{|N|}\sum_{i \in N} Q_{opt}(i)\right) \tag{7}$$

Here, Q_f represents the final fused vector, $\max_{i \in N} Q_{opt}(i)$, $\sum_{i \in N} Q_{opt}(i)$, $\frac{1}{|N|}\sum_{i \in N} Q_{opt}(i)$ represents max pooling, sum pooling and average pooling respectively.

Furthermore, the message-passing mechanism of the employed GCN layer can be defined as,

$$m_K(q_i) = f^M(\sum_{j \in K_(q_i)} f(q_j, e_{i,j})), \tag{8}$$

where $m_K(q_i)$ is the message aggregated from the neighbours $K(q_i)$ of vertex q_i, f^M denotes a differential aggregator function and $f(q_j, e_{ij})$ represents a differential function defining the influence of a neighbouring vertex q_j on q_i through edge feature e_{ij}.

C. Edge Feature Enhancement Module (EFE)

Due to the complex and variable nature of the features extracted from diverse DL models in multiple resolutions, a single-dimensional edge feature is not sufficient to capture the complex dependencies between each feature vector represented as vertices. Specifically, when edge features, e_{ij}, are represented as a scaler $e_{ij}(1)$, the message passing is limited to,

$$m_K(q_i) = f^M\left(\sum_{j \in K_(q_i)} f([q_j(1), \dots, q_j(N)], e_{ij}(1))\right). \tag{9}$$

where f is a differential function defining the influence between neighboring vertices through edge features. To address this limitation, following [35], we introduce task-specific multi-dimensional edge feature, $e_{ij} \in \mathbb{R}^{1 \times D}$, by considering the edge presence, $(\hat{A}_{ij} = 1)$, the corresponding vertex features, q_i, q_j, and the global correlation matrix, Z.

The edge feature enhancement module comprises two blocks: a Vertex-Context Relationship (VCR) modeling block and a Vertex-Vertex Relationship (VVR) modeling block. The functionality of these blocks can be defined as,

1) VCR block: To capture the relationship between vertex features, q_i and q_j, we apply the coss attention between vertex features and the global correlation matrix, Z. Here, the vertex features, q_i and q_j, are treated as queries while the correlation matrix, Z, is treated as keys and values. This process can be expressed as:

$$\begin{aligned} \eta_{i,z} &= VCR(q_i, Z), \\ \eta_{j,z} &= VCR(q_j, Z), \end{aligned} \tag{10}$$

where $\eta_{i,z}$ and $\eta_{j,z}$ represent the vertex-context relationship features captured using the VCR block. The cross attention in the VCR block can be mathematically defined as,

$$VCR(X, Y) = softmax\left(\frac{XW_q\ (YW_k)^\top}{\sqrt{c_k}}\right) YW_v \tag{11}$$

In this expression, W_q,W_k, and W_v are learnable weights corresponding to query, key, and value vectors, and c_k is a scaling factor.

2) VVR block: This is also a cross-attention block in the same form as the VCR block, however, it analyses the output of the VCR block, ($\eta_{i,z}$ and $\eta_{j,z}$), to extract task-specific context features related to adjacent vertices from the generated vertex-context representation. Specifically, it treats $\eta_{i,z}$ as the query and $\eta_{j,z}$ as the key and value to produce a context-aware vertex-vertex relationship, $\eta_{i,z,j}$ and simultaneously, treats $\eta_{j,z}$ as the query and $\eta_{i,z}$ as the key and value to produce $\eta_{j,z,i}$ where:

$$\begin{aligned} \eta_{i,z,j} &= VVR(\eta_{i,z}, \eta_{j,z}) \\ \eta_{j,z,i} &= VVR(\eta_{j,z}, \eta_{i,z}). \end{aligned} \tag{12}$$

Then, to obtain the corresponding multidimensional edge feature vectors, $\hat{e}_{i,j}$ and $\hat{e}_{j,i}$, we flattened $\eta_{i,x,j}$ and $\eta_{j,x,i}$. This operation is denoted as:

$$\begin{aligned} \hat{e}_{i,j} &= f^U(\eta_{i,z,j}) \\ \hat{e}_{j,i} &= f^U(\eta_{j,z,i}). \end{aligned} \tag{13}$$

Finally, the message-passing mechanism can be represented as:

$$m_K(q_i) = f^M\left(\sum_{j \in K_(q_i)} f([q_j(1), .., q_j(N)], \hat{e}_{i,j}\right), \tag{14}$$

where $\hat{e}_{i,j} = [\hat{e}_{i,j}(1), \hat{e}_{i,j}(2), ..., \hat{e}_{i,j}(D)]$ such that the impact on a feature vector from the adjacent vertices is controlled by the N-dimensional task-specific edge features and f is a differential function defining the influence between the neighboring vertices through edge features.

D. Classifier

After obtaining the final fused feature embedding, Q_f, we pass it through a classification layer with softmax activation to predict the retinal disease. This operation can be expressed as,

$$y_n = softmax(f^c(Q_f)), \tag{15}$$

where y_n denotes the predicted disease and f^c represents the classifier.

IV. EVALUATIONS

In this section, we report the results of experiments that we conducted to evaluate and compare the efficiency of the proposed model with existing state-of-the-art methods. We first provide the implementation details (Sec. IV-A), followed by comparisons with the state-of-the-art methods (Sec. IV-B). Ablation evaluations to demonstrate the efficacy of each of the proposed innovations are presented in Sec. IV-C.

A. Implementation Details

The implementation of the proposed model is completed using the python and Pytorch deep learning library. The proposed model was trained for 20 epochs using the Adam optimiser and cosine annealing scheduler with a learning rate of 0.0001. The hyper-parameters K, D, and N were experimentally chosen and were set to 2, 1024, and 8, respectively.

B. Comparisons to Existing State-of-the-art Methods

In this section, we provide comparisons between the proposed method and the existing state-of-the-art (SOTA) methods using three public benchmarks, OCTDL, OCTID, and OCT-2014. The evaluation results are provided in Tab. I. As baseline methods, we have used ResNet50, and VGG16 models used in [36], and the models proposed in [37], [38], [15], [39], [26], [40], [41], and [21]. These models have been selected based on their performance in the respective datasets. To ensure a fair evaluation, we compared the models using five performance parameters: accuracy, F1 score, precision, recall, and AUC.

In the OCTDL dataset, compared to the current SOTA method [38], which achieves an accuracy of 91.1%, the proposed method achieves an accuracy of 94.5%, which is a 4.4% increase. Furthermore, our proposed method demonstrates a competitive F1-score of 90.9%, with higher precision (91.8%) and recall (90.4%) compared to the current SOTA methods.

In the OCTID dataset, our proposed method surpasses all the baselines across all the considered metrics. Specifically, it achieves 6.0%, 5.3%, 4.3%, 5.4%, and 1.1% improvements across accuracy, F1, precision, recall, and AUC scores, respectively, compared to the current SOTA methods. We would like to especially compare the proposed method with [15], which has used both CNN and ViT models for feature extraction at a single resolution and employed a naive fusion approach. We achieve superior robustness compared to this method, demonstrating the utility of the proposed

TABLE I
PERFORMANCE COMPARISON OF BENCHMARK DATASETS WITH EXISTING STATE-OF-THE-ART METHODS. THE BEST RESULTS ARE BOLD AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Refs.	Acc	F1	P	R	AUC
OCTDL	ResNet50 [36]	0.846	0.866	0.898	0.846	**0.988**
	VGG16 [36]	0.859	0.869	0.888	0.859	0.971
	Marciniak [37]	0.798	0.724	0.768	0.708	N/A
	Gupta [38]	0.911	**0.912**	0.901	N/A	0.981
	Proposed	**0.945**	0.909	**0.918**	**0.904**	0.962
OCTID	ResNet50 [36]	0.923	0.927	0.932	0.923	0.979
	VGG16 [36]	0.932	0.927	0.932	0.932	0.970
	Dutta [15]	0.923	0.920	N/A	N/A	N/A
	Singh [39]	0.914	0.912	0.914	0.912	N/A
	Proposed	**0.983**	**0.980**	**0.975**	**0.986**	**0.999**
OCT-2014	Srinivasan [26]	0.956	N/A	N/A	N/A	N/A
	Hussain [40]	0.973	0.972	N/A	N/A	0.990
	Karri [41]	0.846	N/A	0.875	0.819	N/A
	Rasti [21]	N/A	N/A	0.983	0.978	0.990
	Proposed	**0.998**	**0.998**	**0.998**	**0.998**	**1.000**

multi-resolution feature extraction and the efficient graph-based feature fusion strategy. The proposed method achieves similar improvements across all evaluated metrics in the OCT-2014 dataset also outperforming current state-of-the-art methods on these benchmarks.

Overall, these comparisons demonstrate that our proposed multi-resolution and multi-modal feature extraction strategy, combined with effective feature fusion, surpasses the performance of current SOTA models, ensuring accuracy, robustness, and reliability in retinal image classification.

C. Ablation Evaluations

In this section, we illustrate the systematic evaluations conducted to determine the effectiveness of the proposed evaluations using the OCTDL dataset.

*1) **Effectiveness of Multi-Resolution and Multi-Model Feature Extraction**:* To evaluate the effectiveness of our proposed multi-resolution and multi-model feature extraction strategy, we conducted an ablation study by analysing the impact of different pre-existing deep learning models, their ensembles, and the integration of multi-resolution features within a single model. The results are summarized in Tab. II.

First, we assessed the performance of individual models; ResNet50, InceptionV3, and ViT-L/16, which serve as the backbone architectures in our feature extraction pipeline. Among them, ViT-L/16 achieved the highest accuracy of 92%. To leverage the strengths of multiple architectures, we did an experiment by implementing a decision-level ensemble model, which is denoted by Ensemble-I in Tab. II. This approach improved the classification accuracy to 93%, demonstrating improved feature representation compared to individual models. Subsequently, we integrated

TABLE II

COMPARISON OF EFFECTIVENESS OF MULTI-RESOLUTION AND MULTI-MODEL FEATURE EXTRACTION

Model	Acc	F1	P	R	AUC
ResNet50	0.869	0.871	0.886	0.869	0.985
Inception V3	0.879	0.863	0.851	0.879	0.975
ViT-L/16	0.920	0.830	0.909	0.791	0.943
Ensemble - I	0.930	0.89	0.90	0.88	0.98
Ensemble - II	0.901	0.810	0.848	0.785	0.9756
Proposed	**0.945**	**0.909**	**0.918**	**0.904**	**0.962**

multi-resolution feature extraction by aggregating information from different stages of the ResNet50 model and concatenated and passed it to the classifier. This Ensemble-II model achieved an accuracy of 90.1%, which is lower than individual model ensembles. Furthermore, its lower F1 score (82.8%) and recall (80.8%) indicate that while multi-resolution techniques enhance feature diversity, they may require more sophisticated fusion strategies to effectively propagate this information to the classifier, without overloading it with information. We would like to compare this with our proposed method, which integrates both multi-resolution and multi-model feature extraction using graph-based fusion and achieves the highest classification accuracy of 94.5%.

*2) **Effectiveness of Task-Specific Topology Generation and Edge Feature Enhancement**:* To evaluate the impact of Task-Specific Topology Generation and Edge Feature Enhancement on model performance, we conducted several ablation studies. The evaluation results are shown in Tab. III.

Our initial baseline model, which does not use EFE and TTP modules, has achieved 88% accuracy. After integrating the task-specific topology prediction module without the EFE, the accuracy has improved to 93%. Furthermore, the introduction of the EFE module into the baseline model has also contributed to performance improvement. However, the combination of the EFE and TTP modules has outperformed all other models, achieving a 6. 4% improvement compared to the baseline model, with an accuracy of 94. 5%. This result confirms that the combination of task-specific topology and edge feature enhancement allows for a more comprehensive feature integration, leading to an accurate classification performance.

*3) **Effectiveness of Graph-based Feature Fusion**:* In our third ablation study, we compare the effectiveness of different fusion strategies in improving classification performance.

TABLE III

COMPARISON OF EFFECTIVENESS OF TASK-SPECIFIC TOPOLOGY GENERATION AND EDGE FEATURE ENHANCEMENT

Method	Acc	F1	P	R	AUC
Baseline	0.881	0.840	0.808	0.895	0.960
Baseline + EFE	0.930	0.884	0.860	0.9143	0.954
Baseline + TTP	0.930	0.893	0.887	0.903	0.960
Proposed	**0.945**	**0.909**	**0.918**	**0.904**	**0.962**

TABLE IV

COMPARISON OF EFFECTIVENESS OF GRAPH-BASED FEATURE FUSION

Fusion Method	Acc.	F1	P	R	AUC
Naive Fusion	0.871	0.818	0.791	0.866	0.964
Attention-based	0.899	0.857	0.846	0.875	0.965
Decision level	0.928	0.867	0.924	0.832	0.986
Proposed	**0.945**	**0.909**	**0.918**	**0.904**	**0.962**

The results, presented in Tab. IV, demonstrate how various feature fusion approaches impact the accurate classification.

According to the results obtained, the proposed graph-based fusion method outperforms all other fusion methods, achieving the highest values among all the considered metrics. The naive fusion approach, which directly concatenates the extracted multi-resolution and multi-semantic features, yields the lowest accuracy compared to the other methods. This demonstrates the complexity of the extracted features and the need for a more sophisticated fusion strategy. Along these lines, the attention-based fusion method has been able to enhance classification performance, denoting its ability to select informative features to a certain extent. However, the proposed graph-based feature fusion method has been able to elevate this feature refinement to its next level by effectively aggregating the information across multiple models and resolutions, allowing it to achieve the best performance.

Finally, to further validate the generalizability of our proposed approach, we conducted a subject-wise cross-validation evaluation using the OCTDL dataset. The model achieved an accuracy of 0.969, F1-score of 0.953, precision of 0.954, recall of 0.954, and an AUC of 0.980, thereby demonstrating its robust and reliable performance across all key evaluation metrics.

V. CONCLUSIONS

This paper proposed a graph based feature fusion mechanism to integrate multi-modal and multi-resolution feature representations of retinal OCT images for improving the classification performance. The ResNet50, InceptionV3, and ViT models extract multiple semantics from the input at multiple resolutions. A novel graph neural network-based effective feature fusion strategy is proposed to dynamically weight and combine these extracted features based on their importance for the classification task. Extensive experiments were conducted on three public benchmarks: OCTDL, OCTID, and OCT2014, which demonstrated the ability of the proposed framework to outperform the current state-of-the-art algorithms by significant margins, facilitating early and accurate retinal disease diagnosis. We observe two key limitations of the proposed work and suggest the following future research directions: (i) In the present study, we have only investigated the fusion of image features using the proposed graph-based fusion strategy. Future research efforts could be directed to investigate the fusion of diverse modalities, including the textual meta-data about the subjects. Additionally, (ii) hierarchical graph-based structures could be investigated to learn efficient subject-specific representations.

REFERENCES

[1] D. Avitia, J. Baldwin, I. Mcfadden, B. Rowland, P. Rodriguez, and C. Pennington, *Blindness and Visual Impairment*, pp. 3–18. 12 2024.

[2] C. Zhou, S. Li, L. Ye, C. Chen, S. Liu, H. Yang, P. Zhuang, Z. Liu, H. Jiang, J. Han, *et al.*, "Visual impairment and blindness caused by retinal diseases: A nationwide register-based study," *Journal of Global Health*, vol. 13, p. 04126, 2023.

[3] D. Yorston, "Retinal diseases and vision 2020," *Community Eye Health Journal*, vol. 16, 01 2003.

[4] W. Drexler and J. Fujimoto, "State-of-the-art retinal optical coherence tomography," *Progress in retinal and eye research*, vol. 27, pp. 45–88, 02 2008.

[5] L. Balyen and T. Peto, "Promising artificial intelligence-machine learning-deep learning algorithms in ophthalmology," *The Asia-Pacific Journal of Ophthalmology*, vol. 8, no. 3, pp. 264–272, 2019.

[6] Z. L. Teo, Y.-C. Tham, M. Yu, C.-Y. Cheng, T. Y. Wong, and C. Sabanayagam, "Do we have enough ophthalmologists to manage vision-threatening diabetic retinopathy? a global perspective," *Eye*, vol. 34, no. 7, pp. 1255–1261, 2020.

[7] M. Subramanian, K. Shanmugavadivel, O. S. Naren, K. Premkumar, and K. Rankish, "Classification of retinal oct images using deep learning," in *2022 International Conference on Computer Communication and Informatics (ICCCI)*, pp. 1–7, 2022.

[8] A. Choudhary, S. Ahlawat, S. Urooj, N. Pathak, A. Lay-Ekuakille, and N. Sharma, "A deep learning-based framework for retinal disease classification," in *Healthcare*, vol. 11, p. 212, MDPI, 2023.

[9] S. R. Shatil and M. M. J. Kabir, "Retinal oct image classification based on cnn and transfer learning," in *International Conference on Soft Computing and Pattern Recognition*, pp. 433–444, Springer, 2022.

[10] S. Baba, P. Kumari, and P. Saxena, "Retinal disease classification using custom cnn model from oct images," *Procedia Computer Science*, vol. 235, pp. 3142–3152, 2024.

[11] A. Sunija, S. Kar, S. Gayathri, V. P. Gopi, and P. Palanisamy, "Octnet: A lightweight cnn for retinal disease classification from optical coherence tomography images," *Computer methods and programs in biomedicine*, vol. 200, p. 105877, 2021.

[12] P. K. Upadhyay, S. Rastogi, and K. V. Kumar, "Coherent convolution neural network based retinal disease detection using optical coherence tomographic images," *Journal of King Saud University-Computer and Information Sciences*, vol. 34, no. 10, pp. 9688–9695, 2022.

[13] Z. Li, J. Hu, K. Wu, J. Miao, Z. Zhao, and J. Wu, "Local feature acquisition and global context understanding network for very high-resolution land cover classification," *Scientific Reports*, vol. 14, no. 1, p. 12597, 2024.

[14] J. Maurício, I. Domingues, and J. Bernardino, "Comparing vision transformers and convolutional neural networks for image classification: A literature review," *Applied Sciences*, vol. 13, no. 9, p. 5521, 2023.

[15] P. Dutta, K. A. Sathi, M. A. Hossain, and M. A. A. Dewan, "Conv-vit: a convolution and vision transformer-based hybrid feature extraction method for retinal disease detection," *Journal of Imaging*, vol. 9, no. 7, p. 140, 2023.

[16] M. Rahimzadeh and M. R. Mohammadi, "Roct-net: A new ensemble deep convolutional model with improved spatial resolution learning for detecting common diseases from retinal oct images," in *2021 11th International Conference on Computer Engineering and Knowledge (ICCKE)*, pp. 85–91, IEEE, 2021.

[17] J. Wang, H. Peng, S. Chen, and S. Ren, "Less is more: Ensemble learning for retinal disease recognition under limited resources," *arXiv preprint arXiv:2402.09747*, 2024.

[18] K. Pin, J. W. Han, and Y. Nam, "Retinal diseases classification based on hybrid ensemble deep learning and optical coherence tomography images.," *Electronic Research Archive*, vol. 31, no. 8, 2023.

[19] A. Wali, Z. Suhail, S. Naz, and I. Younas, "An ensemble deep learning model for oct image detection and classification," 2024.

[20] G. Naik, N. Narvekar, D. Agarwal, N. Nandanwar, and H. Pande, "Eye disease prediction using ensemble learning and attention on oct scans," in *Future of Information and Communication Conference*, pp. 21–36, Springer, 2024.

[21] R. Rasti, H. Rabbani, A. Mehridehnavi, and F. Hajizadeh, "Macular oct classification using a multi-scale convolutional neural network ensemble," *IEEE transactions on medical imaging*, vol. 37, no. 4, pp. 1024–1034, 2017.

[22] O. Akinniyi, M. M. Rahman, H. S. Sandhu, A. El-Baz, and F. Khalifa, "Multi-stage classification of retinal oct using multi-scale ensemble deep architecture," *Bioengineering*, vol. 10, no. 7, p. 823, 2023.

[23] S. Sotoudeh-Paima, A. Jodeiri, F. Hajizadeh, and H. Soltanian-Zadeh, "Multi-scale convolutional neural network for automated amd classification using retinal oct images," 2022.

[24] M. Kulyabin, A. Zhdanov, A. Nikiforova, A. Stepichev, A. Kuznetsova, V. Borisov, M. Ronkin, A. Bogachev, and S. Korotkich, "Octdl: Optical coherence tomography dataset for image-based deep learning methods," 2023.

[25] P. Gholami and V. Lakshminarayanan, "Optical coherence tomography image retinal database," 2019.

[26] P. P. Srinivasan, L. A. Kim, P. S. Mettu, S. W. Cousins, G. M. Comer, J. A. Izatt, and S. Farsiu, "Fully automated detection of diabetic macular edema and dry age-related macular degeneration from optical coherence tomography images," *Biomedical Optics Express*, vol. 5, pp. 3568–3577, 10 2014.

[27] P. Gholami, P. Roy, M. K. Parthasarathy, and V. Lakshminarayanan, "Octid: Optical coherence tomography image database," *Computers & Electrical Engineering*, vol. 81, p. 106532, 2020.

[28] Y. Pan, J. Liu, Y. Cai, X. Yang, Z. Zhang, H. Long, K. Zhao, X. Yu, C. Zeng, J. Duan, *et al.*, "Fundus image classification using inception v3 and resnet-50 for the early diagnostics of fundus diseases," *Frontiers in Physiology*, vol. 14, p. 1126780, 2023.

[29] T. Fernando, D. Priyasad, S. Sridharan, and C. Fookes, "Decoupled and explainable associative memory for effective knowledge propagation," *IEEE Transactions on Neural Networks and Learning Systems*, 2024.

[30] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.

[31] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2818–2826, 2016.

[32] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021.

[33] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255, Ieee, 2009.

[34] B. Wu, C. Xu, X. Dai, A. Wan, P. Zhang, Z. Yan, M. Tomizuka, J. Gonzalez, K. Keutzer, and P. Vajda, "Visual transformers: Token-based image representation and processing for computer vision," 2020.

[35] X. Liu, J. Lin, and C. Wang, "Graph-based multi-feature fusion method for speech emotion recognition," *arXiv preprint arXiv:2406.07437*, 2024.

[36] M. Kulyabin, A. Zhdanov, A. Nikiforova, A. Stepichev, A. Kuznetsova, M. Ronkin, V. Borisov, A. Bogachev, S. Korotkich, P. A. Constable, and A. Maier, "Octdl: Optical coherence tomography dataset for image-based deep learning methods," *Scientific Data*, vol. 11, Apr. 2024.

[37] T. Marciniak and A. Stankiewicz, "Impact of histogram equalization on the classification of retina lesions from oct b-scans," *Electronics*, vol. 13, p. 4996, 12 2024.

[38] P. Gupta, D. Chakraborty, and D. Guha, "Inter observer variability assessment through ordered weighted belief divergence measure in magdm application to the ensemble classifier feature fusion," 2024.

[39] A. Singh, S. Sengupta, M. A. Rasheed, V. Jayakumar, and V. Lakshminarayanan, "Uncertainty aware and explainable diagnosis of retinal disease," 2021.

[40] M. A. Hussain, A. Bhuiyan, C. D. Luu, R. Theodore Smith, R. H. Guymer, H. Ishikawa, J. S. Schuman, and K. Ramamohanarao, "Classification of healthy and diseased retina using sd-oct imaging and random forest algorithm," *PloS one*, vol. 13, no. 6, p. e0198281, 2018.

[41] S. P. K. Karri, D. Chakraborty, and J. Chatterjee, "Transfer learning based classification of optical coherence tomography images with diabetic macular edema and dry age-related macular degeneration," *Biomedical optics express*, vol. 8, no. 2, pp. 579–592, 2017.